Analyysi I: Limits

Joona Piirainen

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Abstract

Notes for the University of Helsinki course MAT11003, taught in Finnish from 01.08.2026–17.10.2026.

Contents

1 Course information

The official course information is available on the University of Helsinki course page.

2 Lecture notes

2.1 01.09.2026

Definition 2.1. For any real number $x \in \mathbb{R}$, the *absolute value* (or *modulus*) of x, denoted by $|x|$, is defined by

$$|x| = \begin{cases} x, & \text{if } x \geq 0, \\ -x, & \text{if } x < 0. \end{cases}$$

Geometrically, $|x - y|$ is the distance between the points x and y on the real line.

Example 2.2. Fix $y \in \mathbb{R}$ and let $h > 0$. The inequality $|x - y| < h$ can be written in several equivalent ways:

$$\begin{aligned} |x - y| < h &\iff -h < x - y < h \\ &\iff y - h < x < y + h \\ &\iff x \in (y - h, y + h). \end{aligned}$$

Thus, $|x - y| < h$ means that x lies in the open interval of radius h centred at y.

Example 2.3. As a concrete instance of the preceding equivalences,

$$\begin{aligned} |2x-1|<3 &\iff -3<2x-1<3 \\ &\iff -2<2x<4 \\ &\iff -1<x<2 \\ &\iff x\in(-1,2). \end{aligned}$$

Theorem 2.4 (Triangle and reverse triangle inequalities)**.** *For all* $x, y \in \mathbb{R}$,

$$\begin{aligned} |x+y| &\le |x|+|y|, \\ \big||x|-|y|\big| &\le |x-y|. \end{aligned}$$

Proof. By definition 2.1,

$$\begin{aligned} -|x| \le x \le |x|, \\ -|y| \le y \le |y|. \end{aligned}$$

Adding the inequalities gives

$$-\Big(|x|+|y|\Big) \le x+y \le |x|+|y|,$$

which is equivalent to $|x+y| \le |x|+|y|$.

Applying this triangle inequality to $x=(x-y)+y$ gives

$$|x|-|y| \le |x-y|.$$

Interchanging x and y similarly gives $|y|-|x| \le |x-y|$. Therefore

$$-|x-y| \le |x|-|y| \le |x-y|,$$

which is equivalent to the reverse triangle inequality. $\square$